



JSON Fundamentals for SQL Pros





ONE (1) **GIANT TANGENT**
"THERE'S A POINT, I PROMISE."

ACME BRAND





"We choose to go to the moon, and do the other things,
not because they are easy, but because they are hard."
JFK, Rice University, September 1962



Image: NASA

Dr. Wernher von Braun at the Marshall Space Flight Center , October 1967



Image: NASA

Engineers of the Propulsion and Vehicle Engineering Laboratory at work in the Huntsville Industrial Center (HIC) building, February 1964



Alan Shepard in the Rendezvous Docking Simulator,
Langley Research Center, November 1963



Edwin Aldrin Jr. walks on the surface of the moon, July 1969





Space Shuttle Atlantis STS-72, December 1988

Image: NASA

Some real-world benefits
came out of this, too.

Memory Foam



Scratch-resistant lenses



... and freeze-dried foods



There's a space race going on today, too.

There's a space race going on today, too.

Metaphorically speaking.

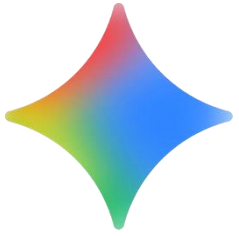
No, not that one.

A photograph of Satya Nadella, CEO of Microsoft, in a thoughtful pose with his hand on his chin. He is wearing glasses and a dark blazer over a blue shirt. The background features the Microsoft logo, which consists of four colored squares (orange, green, blue, and yellow) arranged in a 2x2 grid. The text "It's AI." is overlaid on the image.

It's AI.

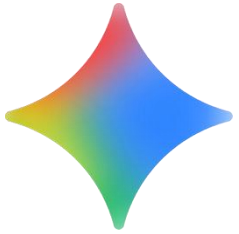


So, how do we talk to AIs?



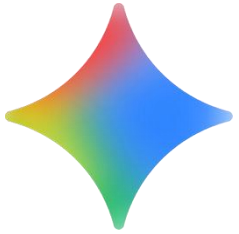
So, how do we talk to AIs?

- AI models expose API endpoints



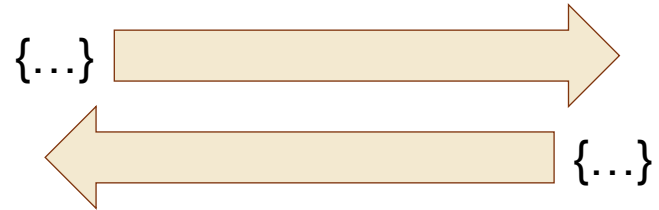
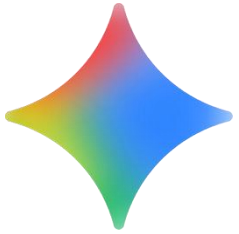
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- AI models expose API endpoints
- APIs use REST over http

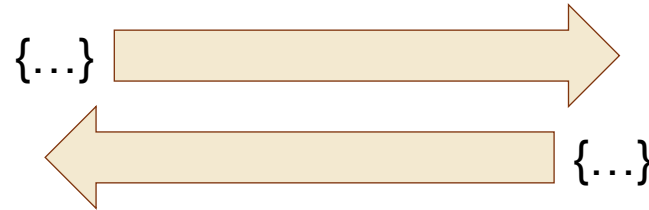
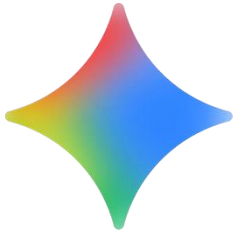


So, how do we talk to AIs?

- AI models expose API endpoints
- APIs use REST over http
- REST responds with JSON



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SQL Server

What's in the box?

INSIDE PLAN OF BOX

1. EX

2.

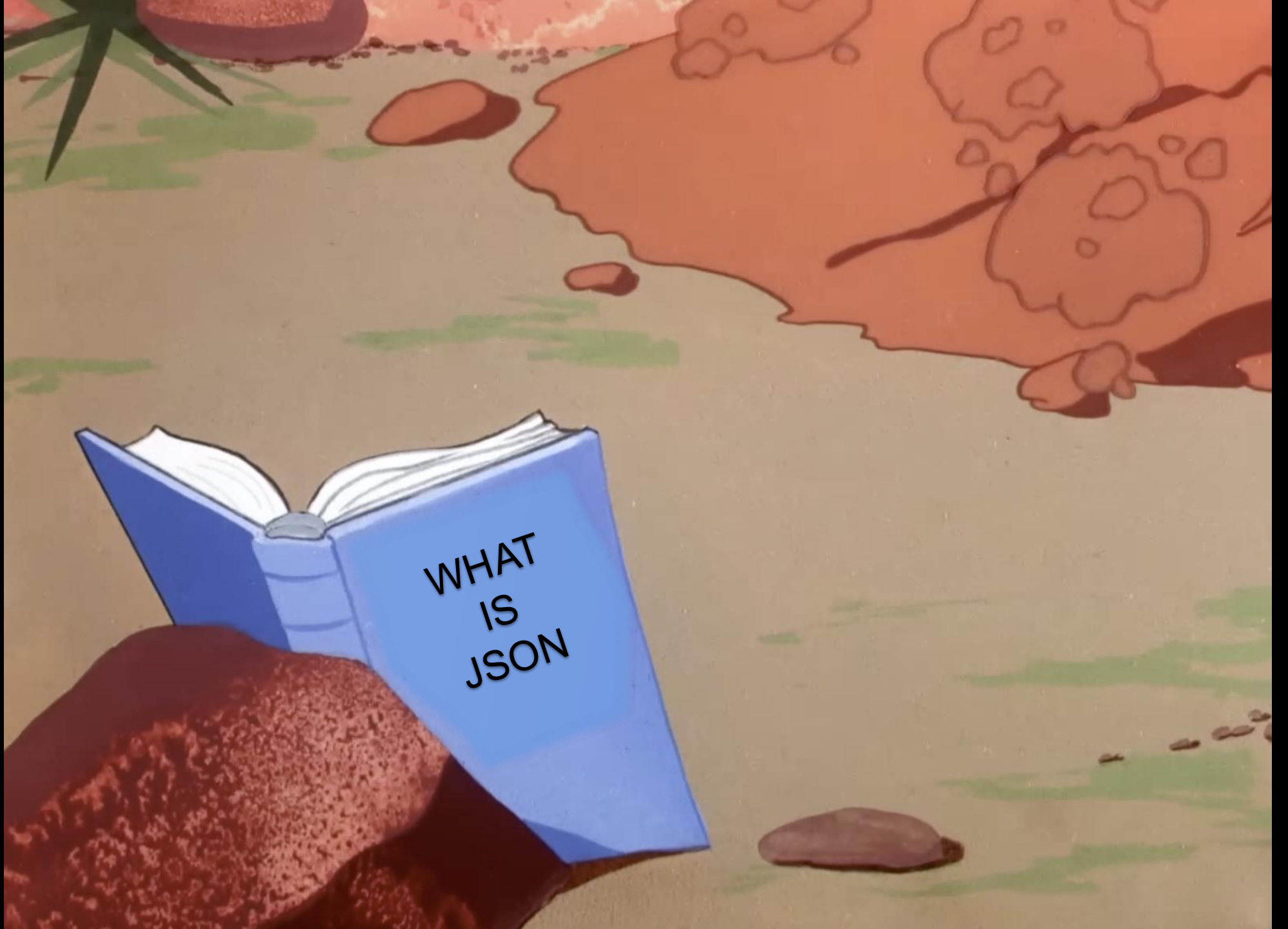
3.

TNT

The diagram shows a top-down view of a rectangular box. Inside the box, there is a large, irregularly shaped mass labeled 'TNT'. To the right of the TNT is a small, rectangular component with a textured surface. A curved line representing a wire or cable connects this component to the TNT. A vertical rod or tube extends from the top of the box down towards the TNT component. Dashed lines with arrows point from labels '1. EX', '2.', and '3.' to various parts of the assembly: '1. EX' points to the top of the vertical rod, '2.' points to a horizontal flange or support on the rod, and '3.' points to the TNT component. A dashed line also points from the bottom right towards the TNT component.

1. EXTERNAL AI MODEL
2. REST SUPPORT
3. "FOR JSON"
4. OPENJSON
5. JSON INDEXES

MICROSOFT CONFIDENTIAL



What is JSON

javascript

```
var event = {};  
  
event.name = "SQLBits 2026";  
event.type = "Conference";  
event.regions = ["Europe", "UK", "Wales"];  
event.venue = "ICC Wales";  
event.startDate = "2026-04-22";  
event.endDate = "2026-04-25";
```

What is JSON

javascript

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var event = {};  
  
event.name = "SQLBits 2026";  
event.type = "Conference";  
event.regions = ["Europe", "UK", "Wales"];  
event.venue = "ICC Wales";  
event.startDate = "2026-04-22";  
event.endDate = "2026-04-25";
```



javascript

```
var event = {  
  "name": "SQLBits 2026",  
  "type": "Conference",  
  "regions": ["Europe", "UK", "Wales"],  
  "venue": "ICC Wales",  
  "startDate": "2026-04-22",  
  "endDate": "2026-04-25"  
};
```

What is JSON

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javascript

```
var event = {  
  "name": "SQLBits 2026",  
  "type": "Conference",  
  "regions": ["Europe", "UK", "Wales"],  
  "venue": "ICC Wales",  
  "startDate": "2026-04-22",  
  "endDate": "2026-04-25"  
};
```



json

"javascript object notation"

```
{  
  "name": "SQLBits 2026",  
  "type": "Conference",  
  "regions": ["Europe", "UK", "Wales"],  
  "venue": "ICC Wales",  
  "startDate": "2026-04-22",  
  "endDate": "2026-04-25"  
}
```


JSON syntax

json

```
{  
  "name": "SQLBits 2026",  
  "type": "Conference"  
}
```

JSON syntax

json

```
{ "name": "SQLBits 2026",  
  "type": "Conference" }
```



json

```
{  
  "name": "SQLBits 2026",  
  "type": "Conference"  
}
```



json

```
{ "name":  
  "SQLBits 2026",  
  "type":  
    "Conference" }
```

JSON syntax

Objects can contain nested objects.

```
json
{
  "name": "SQLBits 2026",
  "type": "Conference",
  "web": {
    "name": "Home",
    "url": "https://sqlbits.com/"
  }
}
```


JSON syntax

Arrays use [and]

json

```
{  
  "name": "SQLBits 2026",  
  "type": "Conference",  
  "web": [  
    { "name": "Home",  
      "url": "https://sqlbits.com/" }  
  ]  
}
```

JSON syntax

json

```
{  
  "name": "SQLBits 2026",  
  "type": "Conference",  
  "web": [  
    { "name": "Home",  
      "url": "https://sqlbits.com/" },  
    { "name": "Agenda",  
      "url": "https://sqlbits.com/attend/the-agenda-2026" }  
  ]  
}
```

JSON syntax

json

```
{
  "name": "SQLBits 2026",
  "type": "Conference",
  "web": [
    { "name": "Home",
      "url": "https://sqlbits.com/" },
    { "name": "Agenda",
      "url": "https://sqlbits.com/attend/the-agenda-2026" },
    { "name": "Sponsors",
      "url": "https://sqlbits.com/sponsor/sponsors/" }
  ]
}
```


JSON syntax

There's no native way to describe dates in JSON.

```
json
{
  "name": "SQLBits 2026",
  "type": "Conference",
  "web": [
    { "name": "Home",
      "url": "https://sqlbits.com/" },
    { "name": "Agenda",
      "url": "https://sqlbits.com/attend/the-agenda-2026" },
    { "name": "Sponsors",
      "url": "https://sqlbits.com/sponsor/sponsors/" }
  ],
  "startDate": "2026-04-22",
  "endDate": "2026-04-25"
}
```

JSON syntax

Numbers are not quoted.

json

```
{
  "name": "SQLBits 2026",
  "type": "Conference",
  "web": [
    { "name": "Home",
      "url": "https://sqlbits.com/" },
    { "name": "Agenda",
      "url": "https://sqlbits.com/attend/the-agenda-2026" },
    { "name": "Sponsors",
      "url": "https://sqlbits.com/sponsor/sponsors/" }
  ],
  "startDate": "2026-04-22",
  "endDate": "2026-04-25",
  "sessionCount": 225
}
```

JSON syntax

true or false can be used for boolean values.

```
json
{
  "name": "SQLBits 2026",
  "type": "Conference",
  "web": [
    { "name": "Home",
      "url": "https://sqlbits.com/" },
    { "name": "Agenda",
      "url": "https://sqlbits.com/attend/the-agenda-2026" },
    { "name": "Sponsors",
      "url": "https://sqlbits.com/sponsor/sponsors/" }
  ],
  "startDate": "2026-04-22",
  "endDate": "2026-04-25",
  "sessionCount": 225,
  "inPerson": true
}
```

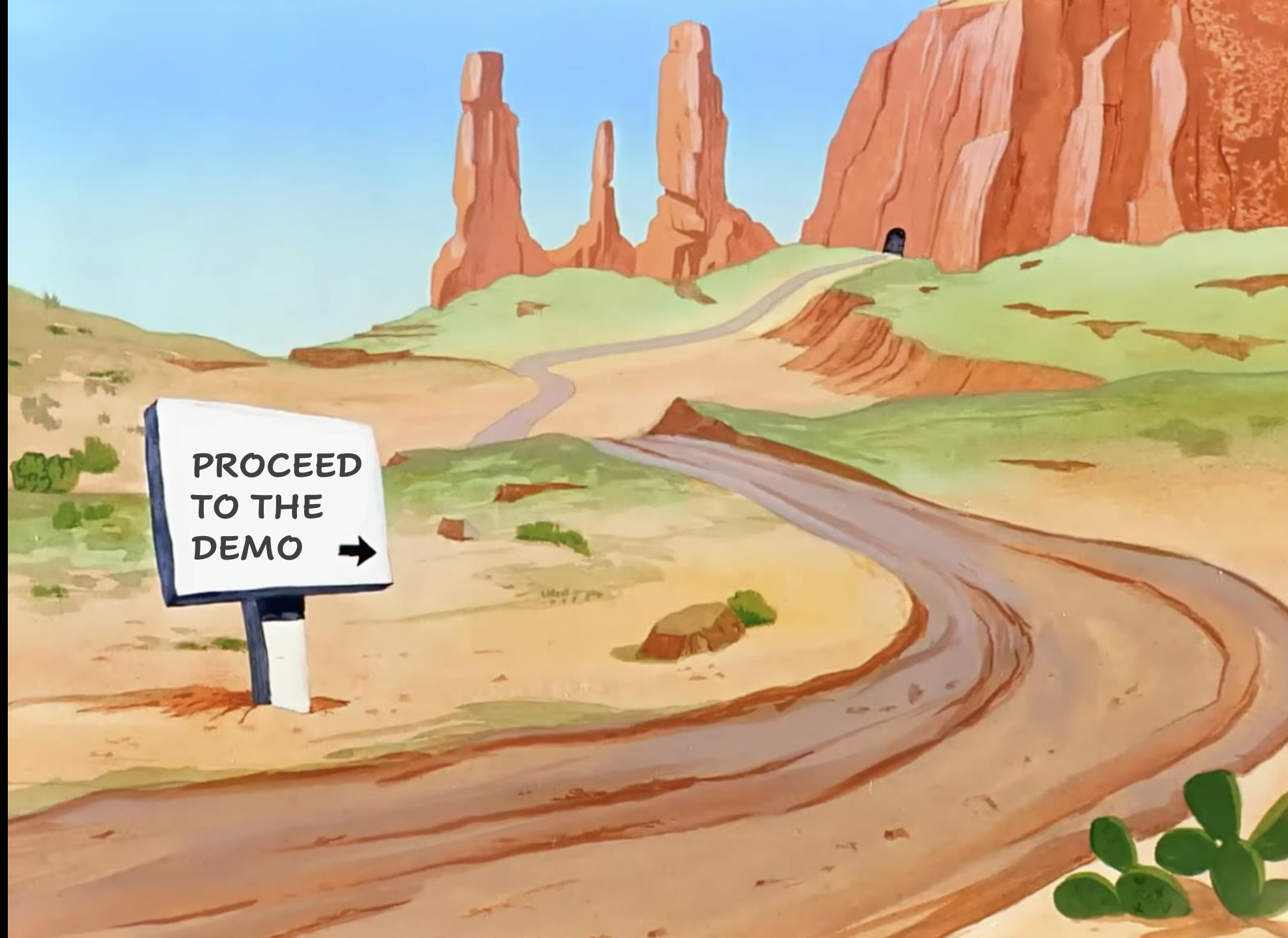
JSON syntax

\ is used as an escape character for ' or ".

\n means line break.

json

```
{
  "name": "SQLBits 2026",
  "type": "Conference",
  "web": [
    { "name": "Home",
      "url": "https://sqlbits.com/" },
    { "name": "Agenda",
      "url": "https://sqlbits.com/attend/the-agenda-2026" },
    { "name": "Sponsors",
      "url": "https://sqlbits.com/sponsor/sponsors/" }
  ],
  "startDate": "2026-04-22",
  "endDate": "2026-04-25",
  "sessionCount": 225,
  "inPerson": true,
  "information": "This year's theme is \"Cartoons\"."
}
```

SQL Server JSON function recap

ISJSON()	Improved in 2022	true/false if a string is valid JSON
JSON_ARRAY()	2022	CONCAT(), but for JSON
JSON_ARRAYAGG()	2025	STRING_AGG(), but returns a JSON array
JSON_MODIFY()		changes individual values in a JSON blob
JSON_OBJECT()	2022	creates a new JSON object
JSON_OBJECTAGG()	2025	merges multiple JSON objects into one object
JSON_PATH_EXISTS()	2022	true/false if a path exists in a JSON object
JSON_QUERY()		returns a filtered subset of a JSON object
JSON_VALUE()		returns a single scalar value from a JSON object
OPENJSON()	Improved in 2022	turns JSON into a table object
JSON_CONTAINS()	2025	like JSON_PATH_EXISTS(), but for values instead of paths
Native json datatype	2025	native, binary storage of JSON data
JSON indexes	2025	similar to XML indexes, allowing for faster lookups
.modify() method	2025	similar to JSON_MODIFY(), used with UPDATE statements



Finally kill your DTS packages

More efficient storage

Faster search



“That’s all Folks!”

<https://github.com/sqlsunday/presentations>

